

Homework 4. Reinforcement Learning

In Lab 4 you will implement several reinforcement learning algorithms, and compare their performance on a scenario from a previous lab. As a preparation for the lab, in this homework you will work with the TD(0) algorithm discussed in class and derive some results that shed some light onto why the algorithm works.

◇

Consider an MDP $(\mathcal{X}, \mathcal{A}, \{\mathbf{P}_a\}, c, \gamma)$ where, for simplicity, we assume that $\mathcal{X} = \{x\}$, i.e., there is a single state. Then, given an arbitrary policy π , the cost-to-go function J^π verifies

$$J^\pi(x) = c_\pi(x) + \gamma J^\pi(x) = \frac{c_\pi(x)}{1 - \gamma}.$$

Let us now consider the TD(0) algorithm for this simple MDP. Given a policy π , the algorithm proceeds by collecting samples (x_t, c_t, x_{t+1}) , used to perform the update

$$J^{(t+1)}(x_t) = J^{(t)}(x_t) + \alpha(c_t + \gamma J^{(t)}(x_{t+1}) - J^{(t)}(x_t)).$$

In our case, $x_t = x_{t+1} = x$, so we can simplify the expression above to

$$J^{(t+1)} = J^{(t)} + \alpha(c_t + (\gamma - 1)J^{(t)}), \quad (1)$$

where we dropped the explicit mention to the state, for simplicity.

We can further rewrite the previous expression as

$$\frac{J^{(t+1)} - J^{(t)}}{\alpha} = c_t + (\gamma - 1)J^{(t)}.$$

For small α , the term on the left hand side resembles the time-derivative of J ,¹ so we could imagine that the trajectories of TD(0) to follow, *in average*, those of the ordinary differential equation (o.d.e.)

$$\dot{J} = \mathbb{E}_\pi [c_t + (\gamma - 1)J]. \quad (2)$$

¹In other words, it measures the variation in the value of J in a time step, dJ/dt .

Exercise 1.

- (a) An *equilibrium point* of an o.d.e.

$$\dot{u}_t = f(u_t)$$

is any point u^* such that $f(u^*) = 0$. Show that J^π is an equilibrium point of the o.d.e. (2).

- (b) Consider now the “energy”

$$E_t = \frac{1}{2}(J^{(t)} - J^\pi)^2.$$

Clearly, the energy E_t is always non-negative, and is only 0 if $J^{(t)} = J^\pi$. Show that, along the trajectories of the o.d.e. (2), the energy always decreases, i.e., that $\dot{E}_t \leq 0$.

- (c) If the trajectories of the TD algorithm in (1) indeed follow, in average, the o.d.e. (2), what do you think the result in (b) suggests about the TD(0)?

Solution 1:

- (a) An equilibrium J_{eq} for the o.d.e. (2) must verify

$$\mathbb{E}_\pi [c_t + (\gamma - 1)J_{eq}] = 0. \quad (3)$$

However, since $\mathbb{E}_\pi [c_t] = c_\pi$, we can rewrite the previous expression as

$$c_\pi + (\gamma - 1)J_{eq} = 0$$

or, equivalently,

$$J_{eq} = \frac{c_\pi}{1 - \gamma}$$

which implies that $J_{eq} = J^\pi$.

- (b) Let us compute $\dot{E}_t$. We have that

$$\begin{aligned} \frac{dE_t}{dt} &= \frac{dE_t}{dJ(x_t)} \cdot \frac{dJ(x_t)}{dt} \\ &= (J^{(t)} - J^\pi)(c_\pi + (\gamma - 1)J^{(t)}). \end{aligned}$$

We now use the fact that

$$c_\pi + (\gamma - 1)J^\pi = 0$$

to get

$$\begin{aligned}\frac{dE_t}{dt} &= (J^{(t)} - J^\pi)(c_\pi + (\gamma - 1)J^{(t)} - c_\pi - (\gamma - 1)J^\pi) \\ &= (\gamma - 1)(J^{(t)} - J^\pi)^2.\end{aligned}$$

Clearly, since $\gamma < 1$, $\gamma - 1 < 0$ and the conclusion follows.

- (c) The result in (b) suggests that the o.d.e. moves to minimize the energy E_t . Since the minimum of such energy is J^π , which is also an equilibrium point of the o.d.e., the previous results suggest that the o.d.e. (and hence the algorithm) converges to J^π .